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Cypher Analytica — Excel for Business and Data Analysis

Level: Beginner to Advanced
Proposed duration: 60 instructor-led hours + 30 hours of independent practice
Schedule: 10 weeks × 3 sessions × 2 hours
Suitable for: Students, job seekers, business owners, and professionals in administration, HR, finance, sales, and operations
Prerequisites: Basic computer knowledge; no prior Excel experience required.

Teaching platform: Excel for Microsoft 365 on Windows. Availability of some functions and tools varies by version; alternative methods should be demonstrated where needed.

Course objectives

By the end of the course, learners should be able to:

- Organize, format, clean, and validate spreadsheet data.
- Build reliable formulas for business calculations.
- Apply lookup, logical, text, and date functions.
- Analyze data using PivotTables and charts.
- Automate repeatable data preparation using Power Query.
- Build an interactive business dashboard.
- Audit workbooks and explain findings clearly.

Module-wise curriculum

Module	Topics covered	Hours
1. Excel Fundamentals and Navigation	Workbooks and worksheets; Ribbon and navigation; entering and editing data; data types; selecting ranges; copying and moving data; Paste Special; AutoFill; keyboard shortcuts; saving in XLSX and CSV formats	3
2. Formatting and Worksheet Management	Cell formatting; alignment; number, currency, percentage, and date formats; custom number formats; row and column management; Freeze Panes; worksheet organization; print areas; page setup	4

Module	Topics covered	Hours
3. Formula Foundations and Cell References	Formula syntax; arithmetic operators; order of operations; relative, absolute, and mixed references; references across sheets; named ranges; SUM, AVERAGE, MIN, MAX, COUNT, and COUNTA	5
4. Logical and Conditional Calculations	IF, nested IF, IFS, AND, OR, NOT; SUMIF, SUMIFS, COUNTIF, COUNTIFS, AVERAGEIF, and AVERAGEIFS; IFERROR; business rules and conditional summaries	5
5. Text, Date, and Time Functions	LEFT, RIGHT, MID, LEN, TRIM, CLEAN, SUBSTITUTE, SEARCH, concatenation, and TEXTJOIN; TODAY, NOW, DATE, YEAR, MONTH, DAY, EOMONTH, NETWORKDAYS, and WORKDAY; elapsed time and date calculations	5
6. Lookup and Dynamic Array Functions	Exact and approximate matching; VLOOKUP, HLOOKUP, INDEX with MATCH, and XLOOKUP; handling missing matches; FILTER, SORT, and UNIQUE; spill ranges; introductory LET for readable formulas	6
7. Data Organization, Cleaning, and Validation	Excel Tables and structured references; sorting and filtering; removing duplicates; Text to Columns; Flash Fill; blank cells; numbers stored as text; inconsistent dates; dropdown lists; validation rules; formula-based conditional formatting	5
8. PivotTables and PivotCharts	Building PivotTables; rows, columns, values, and filters; aggregation choices; grouping dates and numbers; percentage of total; running totals; sorting; slicers; timelines; PivotCharts; refreshing source data	6
9. Charts and Dashboard Design	Column, bar, line, scatter, and combo charts; chart selection; labels and axes; sparklines; KPI summaries; interactive filters; layout and visual hierarchy; avoiding misleading charts	5
10. Power Query for Data Preparation	Importing Excel and CSV files; assigning data types; cleaning and transforming columns; splitting and merging	6

Module	Topics covered	Hours
	columns; append versus merge; unpivoting; grouping; combining files from a folder; refresh workflows	
11. What-If Analysis, Auditing, and Automation Basics	Goal Seek; one-variable and two-variable Data Tables; Scenario Manager overview; tracing precedents and dependents; common formula errors; worksheet protection and its limitations; recording and running a simple macro	4
12. Capstone Project and Assessment	Business requirements; data preparation; formulas and summaries; dashboard construction; validation; documentation; presentation and practical assessment	6
Total	**Guided instruction, practical work, and assessment**	**60**

Practical exercises

Learning area	Suggested task
Formatting and references	Prepare an invoice with quantities, prices, discounts, and totals
Logical functions	Build a student result sheet with grading and eligibility rules
Conditional aggregation	Summarize sales by region, product, and month
Text cleaning	Standardize names, employee codes, and contact information
Date and time calculations	Calculate tenure, working days, deadlines, and attendance hours
Lookups	Retrieve product prices or employee details from a master table
Data validation	Create an entry form with dropdowns and input checks
PivotTables	Analyze department headcount or regional sales performance

Learning area	Suggested task
Power Query	Combine monthly files into a refreshable reporting table
What-if analysis	Find the sales volume needed to achieve a target profit
Dashboards	Build an interactive management report with KPIs and charts

Capstone project options

1. **Sales Performance Dashboard:** Revenue, units sold, target achievement, monthly trends, top products, and regional comparisons.
2. **HR and Attendance Dashboard:** Headcount, department distribution, attendance, leave summaries, and employee tenure.
3. **Inventory Management Workbook:** Opening stock, receipts, issues, closing stock, reorder indicators, and product summaries.
4. **Institute Management Report:** Admissions, batches, attendance, fee collections, outstanding amounts, and course performance.
5. **Budget and Expense Tracker:** Planned versus actual expenses, category summaries, monthly trends, and variance analysis.

Mandatory project deliverables

- Clearly separated source data, calculations, and reporting sheets.
- Structured tables with consistent data types and column names.
- Appropriate formulas, validation rules, and error handling.
- At least one PivotTable and an interactive dashboard.
- A repeatable Power Query transformation.
- Checks reconciling report totals with source data.
- Brief instructions explaining assumptions and refresh steps.

Assessment Structure

Component	Weight
Weekly assignments	25%

Component	Weight
Formula and function practical assessment	20%
Data cleaning and PivotTable assessment	20%
Capstone project	25%
Project presentation and viva	10%
Total	**100%**

Certification Criteria: Minimum 60% overall, submission of mandatory assignments, and successful demonstration of a working capstone.

Interview preparation

Include exercises on absolute references, lookup errors, duplicate detection, conditional calculations, PivotTable refresh issues, numbers stored as text, mixed date formats, and choosing between formulas and Power Query.